

Annotated Bibliography

Raelyn Brooks

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References

- [1] Wenkai Fan. Suspect screening and semi-quantification of organic contaminants in snow and water samples on the qinghai-tibet plateau and high mountain regions. Undergraduate thesis, The College of Wooster, Wooster, OH, 2025. Unpublished thesis.

This unpublished work examines detection and quantification of organic pollutants in high-altitude environments, contributing to understanding contamination in sensitive ecosystems.

- [2] Tobias Kind, Hiroshi Tsugawa, Tomas Cajka, Yan Ma, Zijuan Lai, Sajjan S. Mehta, Gert Wohlgemuth, Dinesh Kumar Barupal, Megan R. Showalter, Masanori Arita, and Oliver Fiehn. Identification of small molecules using accurate mass ms/ms search. *Mass Spectrometry Reviews*, 37(4):513–532, 2018.

This review explains how to identify small molecules using high-resolution MS/MS techniques. It highlights the importance of spectral libraries that include experimental data and computer-generated spectra. The article covers methods for creating in silico spectra using various software tools and developing new databases. It also stresses the need for proper data curation and quality control. From my point of view, it provides valuable insights into how computational approaches and shared spectral resources can aid in analyzing complex samples. These methods are especially helpful for identifying unknown compounds in environmental or natural product research.

- [3] Emma L. Schymanski, Junho Jeon, Rebekka Gulde, Kathrin Fenner, Matthias Ruff, Heinz P. Singer, and Juliane Hollender. Identifying small molecules via high resolution mass spectrometry: Communicating confidence. *Environmental Science & Technology*, 48(4):2097–2098, 2014.

This viewpoint article is essential for non-targeted analysis (NTA) research using HR-MS, such as the computational identification of unknown contaminants in remote environments. It proposes a five-level confidence system for molecular identification, addressing the limitations inherent when reference standards are absent. The framework provides the necessary rigor for reporting the results of

computational tools like molecular networking or cluster analysis. Specifically, Level 3 (Tentative candidate(s)) helps classify structures where computational methods suggest isomers, while Levels 4 and 5 establish clear reporting standards for masses and formulas that serve as critical input or nodes in network-based analysis, enabling transparent data exchange and future traceability of unknown molecules.